GinCol Lab · Aula

Student worksheet

Physics and Chemistry · 1st year of Bachillerato · The atom and the periodic table · Class 4 of 6 · With Abril

A spectroscope at home

Class video: https://youtu.be/83o5UgaC3ZU

The code leads to the video.

Name: _____ Date: _____

The video is in Spanish, with English subtitles. In Spain the decimal separator is a comma: the board writes 546,1 for 546.1 and 6,63 · 10^{-34} for 6.63×10^{-34}. This sheet uses the decimal point.

Before watching the video

1. In the previous class you saw two kinds of spectrum. What spectrum does a very hot solid give? And a glowing low-density gas?

2. If you look at a white screen with a CD, what do you think you will see: a whole rainbow, or only a few colours? Why?

3. A micrometre, or micron (µm), is a thousandth of a millimetre. A hair is about 60 µm thick. How many thousandths of a millimetre is that?

While you watch the video

1. What is recorded on the shiny side of a CD? How far is it from one turn of the track to the next? How many turns fit across the thickness of a hair? And how long would the spiral be if you stretched it out?

2. Why does a CD look coloured? Explain it in your own words: what happens to the little waves that each turn sends back?

3. Why does red come out at a wider angle than violet?

4. What are the parts of a disc spectroscope? Why does the disc have to be in the dark?

5. **The lights in your home.** Complete the table:

Light	Continuous or lines?	Why?
Filament bulb		
LED bulb		
Fluorescent tube		

6. What happens to the filament's spectrum as it heats up little by little?

7. Which element does the fluorescent tube contain? How do we know? Where does the red line that isn't in the table come from?

8. **Your prediction.** If you put yellow on your screen and look at it with a disc, will you see a yellow line, like sodium's? What do you actually see?

9. Why does your eye see yellow even though the screen only lights up green and red?

10. How do you look at sunlight safely? Why can't you point it at the Sun, even if you only look at the disc? How is its spectrum like the filament's, and how is it different?

Practice

1. Three strips, A, B and C: a lithium flame, a fluorescent tube and a filament bulb, in a different order. Which is which? How do you know?

2. Mercury's green line is at 546.1 nm. What energy does each photon carry? ($h = 6.63 \times 10^{-34}$ J·s; $c = 3.00 \times 10^{8}$ m/s)

3. Explain in your own words what makes red leave the disc at a wider angle than violet.

Keep thinking

Your screen fools your eye because it has three colours for your three cones. Many birds have four kinds of cones, and one of them reaches into the ultraviolet. Would your screen fool an eye like that?

If you want to try it (for keen makers)

A cardboard box (a tissue box, a cereal box…) and a CD. At one end, a slit: two strips of cardboard half a millimetre apart. Through a cut in the lid, at the other end, the whole disc, tilted (keep adjusting the tilt until you see the colours). Above the disc, a little window to look through. Cover any other hole. With scissors; the craft knife, with an adult. **Never point it at the Sun, not even if you only look at the disc: it acts as a mirror**. Point it at the sky, at a cloud or at a white wall in the sun; if the light bothers you, look away.

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